

# FOX\_Checklist

**What this is:** every FoxByte feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `FOX_Feature_Implemented.pdf` (same ids).

**Last updated:** 21 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection ( `chore/foxbyte` )

**Source:** `docs/FOX_Checklist.html` → `make feature-doc` → this PDF

☒ Done & tested — automated tests (unit, VM integration or CI)

☒ Done — tested by hand only

☐ Partial — gaps, or not verified end to end

☒ Known issue

☐ Not started

Sub-features: ☒ built ☐ partly ☐ not built yet ☒ broken · "VM §n" = section of `make integration` ; "v2 §n" = `make integration-v2` ; "update" = `make integration-update`

## Where we are

| Area                                    | Features | Done & tested | Done (manual) | Partial | Known issue | Not started | Progress                    |
|-----------------------------------------|----------|---------------|---------------|---------|-------------|-------------|-----------------------------|
| A · Install & setup                     | 9        | 4             | 3             | 2       | ·           | ·           | <div><div></div></div> 78%  |
| B · CLI & services                      | 2        | 2             | ·             | ·       | ·           | ·           | <div><div></div></div> 100% |
| C · Branching & storage                 | 8        | 6             | 1             | 1       | ·           | ·           | <div><div></div></div> 88%  |
| D · Durability                          | 4        | 3             | ·             | 1       | ·           | ·           | <div><div></div></div> 75%  |
| E · Security foundations                | 3        | 3             | ·             | ·       | ·           | ·           | <div><div></div></div> 100% |
| F · SQL gateway                         | 7        | 6             | 1             | ·       | ·           | ·           | <div><div></div></div> 100% |
| G · Accounts & auth                     | 5        | 4             | ·             | 1       | ·           | ·           | <div><div></div></div> 80%  |
| H · Blackbox (record of schema changes) | 18       | 17            | ·             | 1       | ·           | ·           | <div><div></div></div> 94%  |
| I · Import & migration                  | 13       | 5             | 8             | ·       | ·           | ·           | <div><div></div></div> 100% |
| J · ETL pipelines (experimental)        | 4        | 2             | 2             | ·       | ·           | ·           | <div><div></div></div> 100% |
| K · APIs, SDKs & agents                 | 6        | 4             | 2             | ·       | ·           | ·           | <div><div></div></div> 100% |
| L · Web console                         | 11       | 1             | 10            | ·       | ·           | ·           | <div><div></div></div> 100% |
| M · Build, CI & releases                | 6        | 4             | 2             | ·       | ·           | ·           | <div><div></div></div> 100% |
| Total (features built or started)       | 96       | 61            | 29            | 6       | ·           | ·           | <div><div></div></div> 94%  |
| N · Not started (roadmap)               | 11       | ·             | ·             | ·       | ·           | 11          | <div><div></div></div> 0%   |

### Storage engine — not implemented, by design

FoxByte does **not** have its own storage engine. Data is stored by **PostgreSQL 16's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:16-bookworm` image plus wal-g. This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.

What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with wal-g + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

## Suggested next steps

- Run `fox update` against a real release on macOS and Windows (A9) — cut v0.9.0, install it, then update to v0.9.1.
- HA beyond the demo: archive WAL while main runs on the promoted standby, then automatic failover and fencing (D4). (*Safety fixes C3, D4 hazards, H7, I5, I6 done.*)
- Verify downloads everywhere: installers and `fox setup`'s engine download still skip `SHA256SUMS` (A1, A2, A7).

4. ~~Check the web pages that were built but never used signed in: Integrity, Policies, sidebar (L8–L10); add Blackbox filters (L5).~~ Done — both, in the browser.
5. Add automated tests where there are none: imports (I1–I3, I7 — I5 and I8 now covered), SDKs (K3, K4), web UI (L\*).
6. Run the VM integration suites in CI (M2).

## A Install & setup

7 of 9 done

| ID                                                                                            | Feature · sub-features                                                                                                                                                                                                                                                                                                                                          | Tested by                                                                            |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  <b>A1</b>   | <b>One-line installer (macOS / Linux)</b> <ul style="list-style-type: none"> <li>✓ Detects OS / CPU, downloads the release binary</li> <li>✓ Installs to the prefix (sudo only if needed)</li> <li>✓ Stages the Linux engine for the VM on macOS</li> </ul>                                                                                                     | Unit ( <code>test_install_sh.sh</code> , CI); manual                                 |
|  <b>A2</b>   | <b>Windows installer (install.ps1)</b> <ul style="list-style-type: none"> <li>✓ Installs WSL, elevates, resumes after reboot</li> <li>✓ Downloads bb.exe, engine, image build context</li> <li>✓ Ubuntu rootfs verified against Ubuntu's checksums</li> <li>✓ Adds to PATH and runs <code>fox setup</code></li> </ul>                                           | Pester (Windows CI); Windows run 16 Sep                                              |
|  <b>A3</b>   | <b>fox setup on macOS (Lima)</b> <ul style="list-style-type: none"> <li>✓ Creates / starts the Lima VM</li> <li>✓ Installs Docker and ZFS</li> <li>✓ Reinstalls the engine on every run</li> <li>✓ Starts the stack</li> </ul>                                                                                                                                  | Manual (from-scratch VM run)                                                         |
|  <b>A4</b>   | <b>fox setup on Windows (WSL2 + btrfs)</b> <ul style="list-style-type: none"> <li>✓ Imports a dedicated distro with systemd</li> <li>✓ Docker, btrfs and a storage mount unit</li> <li>✓ Quiet steps, detail in install.log</li> <li>✓ Waits for systemd on a cold distro</li> </ul>                                                                            | Unit ( <code>host_wsl_test.go</code> ); Windows run 16 Sep                           |
|  <b>A5</b> | <b>Linux native</b> <ul style="list-style-type: none"> <li>✓ Engine runs in-process, no VM</li> </ul>                                                                                                                                                                                                                                                           | VM integration (all suites run on Linux)                                             |
|  <b>A6</b> | <b>Command forwarding &amp; host file streaming</b> <ul style="list-style-type: none"> <li>✓ macOS → Lima VM, starts the VM if stopped</li> <li>✓ Windows → WSL distro</li> </ul>                                                                                                                                                                               | Unit ( <code>TestImportLocalFile</code> , <code>TestWSLArgs</code> )                 |
|  <b>A7</b> | <b>Always-latest engine binary</b> <ul style="list-style-type: none"> <li>✓ Downloads the newest engine into the cache (ELF check)</li> <li>✓ <code>FOX_NO_REFRESH</code> , <code>FOX_VERSION</code></li> </ul>                                                                                                                                                 | Unit ( <code>host_test.go</code> , <code>update_test.go</code> ); Windows run 16 Sep |
|  <b>A8</b> | <b>New-release notice on fox start</b> <ul style="list-style-type: none"> <li>✓ Background check, 1.5 s limit, printed last</li> <li>✓ Only a promoted, newer, complete release</li> </ul>                                                                                                                                                                      | Unit; update \$1, \$1b, \$2, \$3, \$8; real Mac check                                |
|  <b>A9</b> | <b>fox update</b> <ul style="list-style-type: none"> <li>✓ Downloads and verifies everything before changing anything</li> <li>✓ Stops servers, installs engine, restarts old one on failure</li> <li>✓ <code>up</code> , Blackbox upgrade, <code>start</code> on the new engine</li> <li>✓ Replaces host <code>fox</code> , keeps previous binaries</li> </ul> | Unit; update \$4–\$7; a real Mac run                                                 |

## B CLI & services

2 of 2 done

| ID | Feature · sub-features | Tested by |
|----|------------------------|-----------|
|----|------------------------|-----------|

|   |                                        |                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                   |                                                                             |
|---|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| ✓ | <b>B1 The fox command set</b>          | <ul style="list-style-type: none"> <li>✓ Stack: setup, start, stop, up, down, status, logs, psql, update</li> <li>✓ Branching, backup / restore, HA</li> <li>✓ Blackbox, policy, impact</li> <li>✓ Import, pipelines</li> <li>✓ Users, API keys, admin</li> </ul> | <ul style="list-style-type: none"> <li>✓ <code>fox vm status / fox vm shell</code> (Lima, WSL; Linux says there is no VM)</li> <li>□ <code>branch create --from</code></li> <li>✓ <code>fox uninstall</code> removes it all, retired installs included</li> </ul> | Unit ( <code>main_test.go</code> , <code>update_test.go</code> ); VM suites |
| ✓ | <b>B2 Background services (daemon)</b> | <ul style="list-style-type: none"> <li>✓ Control plane :8080, gateway :6432, agent API :8088</li> <li>✓ Detached, pidfiles and logs in <code>~/ . fox</code></li> </ul>                                                                                           | <ul style="list-style-type: none"> <li>✓ Stop (TERM then KILL), status, logs</li> <li>□ Health check / restart on crash</li> </ul>                                                                                                                                | Unit ( <code>TestStopTerminatesProcess</code> ); VM §1                      |

## C Branching & storage

7 of 8 done

| ID | Feature · sub-features                                                                                                                                                                                                                               | Tested by                                                           |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| ✓  | <b>C1 Storage drivers (ZFS / btrfs)</b> <ul style="list-style-type: none"> <li>✓ ZFS snapshot + clone per branch (macOS VM, Linux)</li> <li>✓ btrfs subvolume snapshots (Windows)</li> <li>✓ Pool / filesystem created on a loopback file</li> </ul> | Unit; VM §2, §2b (ZFS); btrfs manual on Windows                     |
| ✓  | <b>C2 Branch lifecycle</b> <ul style="list-style-type: none"> <li>✓ Create, list, delete</li> <li>✓ Reset to parent</li> <li>✓ Suspend / resume</li> <li>✓ Recreate a container from its surviving data</li> </ul>                                   | Unit; VM §2, §2b, §4                                                |
| ✓  | <b>C3 Protection of main</b> <ul style="list-style-type: none"> <li>✓ Delete and reset refused</li> </ul>                                                                                                                                            | Unit; VM §4, §6                                                     |
| ✓  | <b>C4 Private networking (no host ports)</b> <ul style="list-style-type: none"> <li>✓ Branch databases reachable only through the gateway</li> </ul>                                                                                                 | Unit ( <code>TestParsePublishedPort</code> , <code>TestDSN</code> ) |
| ✓  | <b>C5 Scale-to-zero &amp; wake on connect</b> <ul style="list-style-type: none"> <li>✓ Idle branches suspended automatically</li> <li>✓ Next connection wakes the branch (~0.3–0.7 s)</li> </ul>                                                     | VM §7                                                               |
| 🔗  | <b>C6 Storage quotas</b> <ul style="list-style-type: none"> <li>✓ Space reserved for main (ZFS)</li> <li>🔗 Per-branch cap (opt-in)</li> </ul>                                                                                                        | Manual                                                              |
| ✓  | <b>C7 Capacity &amp; status</b> <ul style="list-style-type: none"> <li>✓ Services, main health, backups, branches</li> <li>✓ Storage used / available</li> </ul>                                                                                     | VM §1                                                               |
| ✓  | <b>C8 Provisioning &amp; image pull</b> <ul style="list-style-type: none"> <li>✓ Docker, storage pool and image prepared on up</li> </ul>                                                                                                            | Manual                                                              |

## D Durability

3 of 4 done

| ID | Feature · sub-features                                                                                                                                                     | Tested by      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| ✓  | <b>D1 WAL archiving to object storage</b> <ul style="list-style-type: none"> <li>✓ wal-g ships main's WAL to MinIO, off the commit path</li> </ul>                         | VM §3          |
| ✓  | <b>D2 Base backups</b> <ul style="list-style-type: none"> <li>✓ Create and list</li> <li>✓ Follow the primary after a failover; listing works with main stopped</li> </ul> | VM §3, §3b, §6 |

|   |                                     |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                               |                              |
|---|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| ✓ | <b>D3 Point-in-time restore</b>     | <ul style="list-style-type: none"> <li>✓ Restore to a timestamp or latest on :5433</li> <li>✓ Per-install credentials</li> <li>✓ Starts from the newest base backup that precedes the target</li> </ul>                                                   | <ul style="list-style-type: none"> <li>✓ A point no backup precedes is refused, naming the oldest</li> </ul>                                                                                                                                  | Unit; VM §3, §3b, §6; v2 §1a |
| 🔥 | <b>D4 HA standby &amp; failover</b> | <ul style="list-style-type: none"> <li>✓ Streaming hot standby</li> <li>✓ Manual failover, <code>main</code> rerouted</li> <li>✓ Nothing deletes or stops the promoted standby</li> <li>✓ <code>fox up</code> brings it back, not the old main</li> </ul> | <ul style="list-style-type: none"> <li>✓ <code>fox ha failback</code>, keeping every write</li> <li>✓ The promoted standby archives WAL, so backups and PITR cover the failover</li> <li>❑ Automatic failover, fencing, multi-host</li> </ul> | Unit; VM §6, §7              |

## E Security foundations 3 of 3 done

| ID   | Feature · sub-features                                                                                                                                                                                                                                                                                    | Tested by                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| ✓ E1 | <b>Per-install generated secrets</b> <ul style="list-style-type: none"> <li>✓ Generated Postgres and MinIO passwords</li> <li>✓ Used by restore too</li> </ul>                                                                                                                                            | Unit (indirect); v2 §1a                             |
| ✓ E2 | <b>TLS everywhere</b> <ul style="list-style-type: none"> <li>✓ Self-signed TLS on all three services, zero setup</li> <li>🔥 TLS optional at the gateway (<code>sslmode=disable</code> still connects)</li> </ul>                                                                                          | Unit<br>( <code>TestReadStartupUpgradesTLS</code> ) |
| ✓ E3 | <b>Keys are made by a person, not by start</b> <ul style="list-style-type: none"> <li>✓ Start creates no account and no key, and caches none on disk</li> <li>✓ The banner shows the connection string and how to make a key</li> <li>✓ The first account created gets the admin grant on main</li> </ul> | Integration §1                                      |

## F SQL gateway 7 of 7 done

| ID   | Feature · sub-features                                                                                                                                                                           | Tested by                           |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| ✓ F1 | <b>Routing by database name</b> <ul style="list-style-type: none"> <li>✓ One endpoint :6432, database name = branch</li> <li>❑ Connection pooling, rate limiting</li> </ul>                      | Unit; VM §1, §6                     |
| ✓ F2 | <b>API-key authentication</b> <ul style="list-style-type: none"> <li>✓ <code>key_</code> key as the Postgres password</li> <li>✓ Bad key rejected with a Postgres error</li> </ul>               | VM §1                               |
| ✓ F3 | <b>Backend login (SCRAM)</b> <ul style="list-style-type: none"> <li>✓ Cleartext, md5, SCRAM-SHA-256</li> <li>✓ Clients never see the database password</li> <li>❑ Unit test for SCRAM</li> </ul> | VM §1, §6, §8 (exercised)           |
| ✓ F4 | <b>Per-user Postgres login roles</b> <ul style="list-style-type: none"> <li>✓ Role per user, actor can't be forged</li> <li>✓ Permissions of non-superuser <code>db_client</code></li> </ul>     | Manual                              |
| ✓ F5 | <b>Attribution session settings</b> <ul style="list-style-type: none"> <li>✓ Actor kind, branch, session id per connection</li> <li>🔥 Kind / branch / session client-settable</li> </ul>         | VM §8                               |
| ✓ F6 | <b>Ctrl-C (CancelRequest) forwarding</b> <ul style="list-style-type: none"> <li>✓ Cancel a running query through the gateway</li> </ul>                                                          | Unit                                |
| ✓ F7 | <b>Errors &amp; insecure build tag</b> <ul style="list-style-type: none"> <li>✓ Real Postgres error responses</li> <li>✓ No-auth bypass impossible in release builds</li> </ul>                  | Unit ( <code>TestSendError</code> ) |

## G Accounts & auth 4 of 5 done

| ID   | Feature · sub-features                                                                                                                                                                                                                                                                             | Tested by                                                                                   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| ✓ G1 | <b>Auth store (SQLite)</b> <ul style="list-style-type: none"> <li>✓ Users, keys, sessions, OAuth ids, pipelines</li> <li>✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once</li> <li>❑ Roles / permissions, multi-tenant projects</li> </ul> | Unit ( <code>TestVerifyKeyConcurrent</code> , <code>TestConcurrentFirstOpen</code> ); VM §1 |

|   |                                                                                                                                                           |                                                                                        |                       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------|
| ✓ | <b>G2 Email &amp; password users</b><br>✓ Create account, log in<br>□ Password change / reset                                                             | □ List and delete users                                                                | Unit; VM §1           |
| ✓ | <b>G3 API keys</b><br>✓ Create (shown once), list, revoke<br>✓ Bearer / X-API-Key and gateway password                                                    | ○ Revoking an unknown id reports success                                               | Unit; VM §1           |
| ✓ | <b>G4 Web sessions &amp; signup control</b><br>✓ httpOnly session cookie                                                                                  | ✓ Signup can be closed                                                                 | Unit ( TestSessions ) |
| ○ | <b>G5 OAuth (GitHub / Google)</b><br>✓ Sign-in flows built<br>✓ Defaults match what the engine serves ( https://localhost:8080 ; web origin = public URL) | ✓ Session cookie Secure / SameSite follow the origin<br>□ Tested with a real OAuth app | Unit; VM §1b          |

## H Blackbox (record of schema changes)

17 of 18 done

| ID | Feature · sub-features                                                                                                                                                       | Tested by                                                                                                                                                  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓  | <b>H1 DDL recording</b><br>✓ Who, tool, branch, statement, status, risk<br>✓ Inherited by every branch                                                                       | VM §8<br>□ Role / database changes, some dropped object types                                                                                              |
| ○  | <b>H2 Hash chain, append-only &amp; verify</b><br>✓ Rows can't be edited or deleted normally<br>✓ fox ledger verify<br>○ Superuser rewrite caught only via anchors (H10–H12) | VM §8; v2 §0, §6b<br>✓ bb.policy changes recorded (append-only, not chained)                                                                               |
| ✓  | <b>H3 Attribution</b><br>✓ Humans via their login role<br>✓ Agents via their own agent-<id> role                                                                             | Unit; VM §8; v2 §1c<br>✓ Kind derived from the role, so neither can pose as the other<br>✓ Tool and session are client-declared by design, and labelled so |
| ✓  | <b>H4 Destructive-DDL guardrail</b><br>✓ DROP TABLE / DROP SCHEMA refused, attempt recorded<br>✓ Override only superusers / db_admin , use recorded                          | VM §8; v2 §1b, §6b<br>✓ "flag" and "allow" actions; unknown actions refused<br>✓ No hang when blocked after a write in the same transaction                |
| ✓  | <b>H5 Risk flags</b><br>✓ Type change, drop column flagged<br>✓ Judged on the running statement (no comments or literals)                                                    | Unit; v2 §6b<br>✓ Automated test                                                                                                                           |
| ✓  | <b>H6 Blackbox access (CLI / REST / MCP / UI / SDK)</b><br>✓ CLI, REST with filters, MCP, web page, SDKs                                                                     | v2 §0, §5<br>□ Per-user access control                                                                                                                     |
| ✓  | <b>H7 fox ledger revert --to</b><br>✓ Help and output say it's a point-in-time restore of main                                                                               | Unit (help text); VM §3 (the restore)<br>✓ Points to H13 for a branch from before a change                                                                 |
| ✓  | <b>H8 Override permission (db_admin) &amp; upgrade</b><br>✓ fox admin grant revoke list<br>✓ First user granted automatically                                                | v2 §1b, §6<br>✓ fox blackbox upgrade [--all]<br>✓ Grant / revoke from the web (admins only)                                                                |
| ✓  | <b>H9 Blackbox 2.0 capture</b><br>✓ Transaction id and WAL position per entry<br>✓ Task, tool-call hash, override use                                                        | Unit; v2 §2<br>✓ Fail-safe, append-only<br>✓ Kill switch not settable by clients                                                                           |
| ✓  | <b>H10 Checkpoints &amp; anchors</b><br>✓ Merkle root in the database and in read-only anchor files<br>✓ On a schedule and on demand                                         | Unit; v2 §3, §3b<br>✓ Published anchor format<br>□ Cloud write-once anchors (S3 Object Lock)                                                               |
| ✓  | <b>H11 fox-verify (independent verifier)</b><br>✓ Checks a live branch or an export                                                                                          | Unit; v2 §3<br>✓ Release binaries for every platform                                                                                                       |

|   |                                                     |                                                                                                                                                                                                       |                                                                                                                                                                                       |                  |
|---|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| ✓ | <b>H12 Blackbox integrity &amp; export</b>          | <ul style="list-style-type: none"> <li>✓ INTACT / TAMPERED via CLI, REST, MCP</li> <li>✓ JSON-lines export</li> </ul>                                                                                 | <input type="checkbox"/> Streaming for very large ledgers                                                                                                                             | v2 §3            |
| ✓ | <b>H13 Branch from just before a Blackbox entry</b> | <ul style="list-style-type: none"> <li>✓ Restore + replay up to the change's transaction</li> <li>✓ By transaction id or timestamp</li> </ul>                                                         | <ul style="list-style-type: none"> <li>✓ CLI, REST, MCP, web panel</li> <li>✓ main untouched</li> </ul>                                                                               | Unit; v2 §4      |
| ✓ | <b>H14 Blackbox names &amp; aliases</b>             | <ul style="list-style-type: none"> <li>✓ <code>fox blackbox</code>, <code>/blackbox</code> routes, MCP and SDK names</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>✓ Every original name still works</li> </ul>                                                                                                   | Unit; v2 §5      |
| ✓ | <b>H15 Policy gate (warn / block)</b>               | <ul style="list-style-type: none"> <li>✓ BBX02 warn / BBX01 block with JSON detail</li> <li>✓ Blocked attempts recorded</li> <li>✓ Per-rule <code>db_admin</code> override</li> </ul>                 | <ul style="list-style-type: none"> <li>✓ CLI, REST, MCP, web page; custom rules</li> <li>✓ Patterns match the running statement</li> <li><input type="checkbox"/> TRUNCATE</li> </ul> | Unit; v2 §6, §6b |
| ✓ | <b>H16 Agent provenance &amp; execute_change</b>    | <ul style="list-style-type: none"> <li>✓ Session, task, parent session on every agent change (also through the Gateway DSN)</li> <li>✓ MCP <code>execute_change</code> with policy preview</li> </ul> | <ul style="list-style-type: none"> <li>✓ <code>fox blackbox sessions</code>, REST</li> <li>🟡 Provenance is what the agent says (not verified)</li> </ul>                              | Unit; v2 §7      |
| ✓ | <b>H17 Impact analysis</b>                          | <ul style="list-style-type: none"> <li>✓ Dependents (views, foreign keys, indexes, depth)</li> <li>✓ Other branches that have the object</li> <li>✓ Policy verdict and risk score</li> </ul>          | <ul style="list-style-type: none"> <li>✓ CLI, REST, MCP, inside <code>execute_change</code></li> <li>🟡 Functions, types, schemas not recognised from SQL</li> </ul>                   | Unit; v2 §8      |
| ✓ | <b>H18 Blackbox diff between branches</b>           | <ul style="list-style-type: none"> <li>✓ Fork point, changes only on each side</li> <li>✓ Objects changed on both</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>✓ CLI, REST, MCP</li> <li><input type="checkbox"/> Merge (diff only today)</li> </ul>                                                          | Unit; v2 §8      |

## I Import & migration

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| ID   | Feature · sub-features                                                                                                                                                            | Tested by         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ✓ I1 | <b>From Postgres</b> <ul style="list-style-type: none"> <li>✓ Schema and data via <code>pg_dump</code></li> <li>✓ Zero-table result reported as an error</li> </ul>               | Manual            |
| ✓ I2 | <b>From MySQL ≤5.7 / MariaDB</b> <ul style="list-style-type: none"> <li>✓ <code>pgloader</code> with type mapping (multi-arch image)</li> </ul>                                   | Manual            |
| ✓ I3 | <b>From MySQL 8.x</b> <ul style="list-style-type: none"> <li>✓ Native <code>mysqldump</code> path (<code>cached_sha2</code> auth)</li> <li>✓ Types, NULLs, unicode, PK</li> </ul> | Manual            |
| ✓ I4 | <b>From MongoDB</b> <ul style="list-style-type: none"> <li>✓ Collection → typed table, exact name</li> </ul>                                                                      | VM §9, §10        |
| ✓ I5 | <b>From a .sql dump</b> <ul style="list-style-type: none"> <li>✓ Runs the dump into a new branch</li> <li>✓ Fails listing every failed statement (instance kept)</li> </ul>       | Unit; VM §11      |
| ✓ I6 | <b>From JSON / NDJSON</b> <ul style="list-style-type: none"> <li>✓ Columns inferred from top-level keys</li> </ul>                                                                | Unit; VM §10, §11 |
| ✓ I7 | <b>From CSV</b> <ul style="list-style-type: none"> <li>✓ Table from the header, exact column names</li> </ul>                                                                     | Manual            |

|   |            |                                                                                                                                                                                     |                                                                                                       |              |
|---|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------|
| ✓ | <b>I8</b>  | <b>Continuous replication &amp; cutover</b><br>✓ Initial copy + streaming (logical replication)<br>✓ Cutover keeps data (CLI)<br>✓ Status and cutover from REST and the Import page | ✓ A replicating branch is never auto-suspended<br>✓ Automated test (VM §5b) and driven in the browser | Unit; VM §5b |
| ✓ | <b>I9</b>  | <b>Guard handling during import</b><br>✓ Guardrail paused, recording stays on                                                                                                       | ✓ Re-armed even after a crash                                                                         | Manual       |
| ✓ | <b>I10</b> | <b>Live progress (CLI &amp; SSE)</b><br>✓ Live log and progress bar                                                                                                                 | ⚠️ psql subprocess output not streamed                                                                | Manual       |
| ✓ | <b>I11</b> | <b>Import REST endpoints</b><br>✓ From a connection string or an uploaded file                                                                                                      | <input type="checkbox"/> Target name validation                                                       | Manual       |
| ✓ | <b>I12</b> | <b>Web Import page</b><br>✓ Database and file panels with live console                                                                                                              | ⚠️ Mentions a SQLite loader that doesn't exist                                                        | Manual       |
| ✓ | <b>I13</b> | <b>Exact identifier fidelity</b><br>✓ Case kept ( <code>userId</code> , <code>leaseAiChats</code> )                                                                                 | ⚠️ Field order of JSON sources not kept                                                               | Unit; VM §10 |

## J ETL pipelines (experimental)

4 of 4 done

| ID | Feature · sub-features                                                                                                                                                 | Tested by   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| ✓  | <b>J1</b> <b>Pipeline engine</b><br>✓ Extract → raw schema → SQL models → tests, in a fresh branch<br>✓ <code>ref()</code> / <code>source()</code> templating          | Unit; VM §9 |
| ✓  | <b>J2</b> <b>Data-quality tests</b><br>✓ <code>not_null</code> , <code>unique</code> , <code>accepted_values</code> , <code>row_count_min</code> , <code>custom</code> | Unit; VM §9 |
| ✓  | <b>J3</b> <b>Stored pipelines &amp; run history</b><br>✓ Saved per user, run over the API, history<br><input type="checkbox"/> Read a run's log                        | Manual      |
| ✓  | <b>J4</b> <b>Pipeline editor (web)</b><br>✓ Create, edit, run, results, history                                                                                        | Manual      |

## K APIs, SDKs & agents

6 of 6 done

| ID | Feature · sub-features                                                                                                                                                                         | Tested by               |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| ✓  | <b>K1</b> <b>Control-plane REST API</b><br>✓ Branches, query, Blackbox, policy, impact, import, pipelines, keys                                                                                | Unit; VM §1; v2 §0–§8   |
| ✓  | <b>K2</b> <b>OpenAPI spec</b><br>✓ Served, Blackbox 2.0 endpoints documented                                                                                                                   | Unit (drift check)      |
| ✓  | <b>K3</b> <b>Python SDK</b><br>✓ Branches, query, Blackbox<br><input type="checkbox"/> Import, pipelines, keys, agents                                                                         | Manual                  |
| ✓  | <b>K4</b> <b>TypeScript SDK</b><br>✓ Same client for Node and browsers<br><input type="checkbox"/> Tests                                                                                       | Manual                  |
| ✓  | <b>K5</b> <b>Agent Branch API</b><br>✓ A branch per agent, own non-superuser role<br>✓ Provenance (task, parent, session)<br>✓ DSN reachable from the host (gateway-routed, branch-scoped key) | Unit; VM §5; v2 §1c, §7 |



|   |                                                                                                                                                                                                                                   |                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| ✓ | <b>K6 MCP server</b>                                                                                                                                                                                                              | Unit; v2 §1c–§8                                                                                       |
| ✓ | <ul style="list-style-type: none"> <li>Branches, run_sql, Blackbox tools</li> <li>policy_check, execute_change, impact, diff</li> <li>Requires an API key and acts as that account; a scoped key is held to its branch</li> </ul> | <ul style="list-style-type: none"> <li>Client config &amp; 16-tool reference (docs/mcp.md)</li> </ul> |

L

Web console

11 of 11 done

| ID    | Feature · sub-features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Tested by                                               |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| ✓ L1  | <b>Embedded single-origin UI</b> <ul style="list-style-type: none"> <li>✓ Served by the engine at :8080</li> <li>□ Automated UI tests (whole area)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      | Manual                                                  |
| ✓ L2  | <b>Login &amp; signup</b> <ul style="list-style-type: none"> <li>✓ Log in / sign up tabs, OAuth buttons</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Manual                                                  |
| ✓ L3  | <b>Dashboard</b> <ul style="list-style-type: none"> <li>✓ Live tiles</li> <li>✓ Branch table: create, suspend, resume, delete, copy DSN</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                 | Manual                                                  |
| ✓ L4  | <b>SQL console &amp; browser</b> <ul style="list-style-type: none"> <li>✓ Schema sidebar, rows, structure, indexes</li> <li>✓ Values shown as Postgres writes them (uuid, interval, inet, bytea)</li> <li>✓ SQL editor</li> <li>✓ System tables hidden by default, with a show button</li> <li>✓ Runs as the signed-in user; one-run override switch</li> <li>✓ Per-rule override for policy blocks</li> </ul>                                                                                                                                                                     | Manual; API in v2 §6                                    |
| ✓ L5  | <b>Blackbox page</b> <ul style="list-style-type: none"> <li>✓ Entries per branch with badges</li> <li>✓ Table / risk / date filters, and "Clear filters"</li> <li>✓ Verify button; a row expands to its statement and session</li> </ul>                                                                                                                                                                                                                                                                                                                                           | v2 §5b; browser                                         |
| ✓ L6  | <b>API keys page</b> <ul style="list-style-type: none"> <li>✓ Create, copy, list, revoke</li> <li>✓ A new key is shown inside its connection string (E3's next step)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                    | Manual; browser review 21 Sep                           |
| ✓ L7  | <b>Landing, Guide &amp; Docs pages</b> <ul style="list-style-type: none"> <li>✓ Landing, developer guide, reference</li> <li>✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                | Type-check + build                                      |
| ✓ L8  | <b>Blackbox integrity page</b> <ul style="list-style-type: none"> <li>✓ Verify, checkpoint, export, branch-before</li> <li>✓ Used signed in; every result matched the CLI</li> <li>○ Entry list needs a refresh to show a new change</li> </ul>                                                                                                                                                                                                                                                                                                                                    | Type-check + build; browser review 16 Sep               |
| ✓ L9  | <b>Policies page</b> <ul style="list-style-type: none"> <li>✓ Rules, warn/block, custom rules, preview, evaluations</li> <li>✓ Used signed in; matched the CLI, wrote only to the chosen branch</li> <li>✓ Non-admin change refused with a readable message</li> <li>✓ Who can override: list, grant, revoke</li> </ul>                                                                                                                                                                                                                                                            | Type-check + build; API in v2 §6; browser review 16 Sep |
| ✓ L10 | <b>App sidebar layout</b> <ul style="list-style-type: none"> <li>✓ Every link opens its page, highlighted, with breadcrumb</li> <li>✓ Collapse persists across a reload; theme toggle persists</li> <li>✓ No console errors on any page</li> <li>□ Under-860px drawer (needs a manual look)</li> </ul>                                                                                                                                                                                                                                                                             | Type-check + build; browser review 16 Sep               |
| ✓ L11 | <b>Design system &amp; FoxByte theme</b> <ul style="list-style-type: none"> <li>✓ Tokens for colour, type, space, radius, elevation and motion; components use no raw values</li> <li>✓ Ember palette in both themes; filled controls legible ( --accent-ink )</li> <li>✓ Fox mark from theme variables, wordmark from brand.json, favicon</li> <li>✓ Console, Blackbox, Policies and API keys re-laid out; one table style</li> <li>✓ Every page reviewed in both themes; console checked at 375px</li> <li>□ Not yet reviewed against a running engine with real data</li> </ul> | Type-check + build; browser review 21 Sep               |

M

Build, CI & releases

6 of 6 done

| ID | Feature · sub-features | Tested by |
|----|------------------------|-----------|
|----|------------------------|-----------|

|   |                                       |                                                                                                                                                                                                         |                                                                                                                                                                               |                                               |
|---|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| ✓ | <b>M1 Makefile targets</b>            | <ul style="list-style-type: none"> <li>✓ build, vet, test, integration suites</li> <li>✓ Suites run only in a throwaway test VM (guarded)</li> </ul>                                                    | <ul style="list-style-type: none"> <li>✓ release, web, WSL builds, feature-doc (both PDFs)</li> </ul>                                                                         | CI                                            |
| ✓ | <b>M2 Linux CI</b>                    | <ul style="list-style-type: none"> <li>✓ Build, vet, test, gofmt</li> <li>✓ Release script tests, checklist sync check</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>□ VM integration suites in CI</li> </ul>                                                                                               | CI                                            |
| ✓ | <b>M3 Windows CI</b>                  | <ul style="list-style-type: none"> <li>✓ Build, vet, test, Pester, installer encoding</li> <li>✓ Guard tests read backslash paths, and generated files are pinned to LF (21 Sep)</li> </ul>             | <ul style="list-style-type: none"> <li>□ End-to-end WSL install test</li> </ul>                                                                                               | CI                                            |
| ✓ | <b>M4 Releases &amp; GHCR image</b>   | <ul style="list-style-type: none"> <li>✓ Binaries for 5 platforms + fox-verify</li> <li>✓ SHA256SUMS merged safely between jobs</li> </ul>                                                              | <ul style="list-style-type: none"> <li>✓ Fails unless fox update 's files are published</li> <li>✓ Multi-arch Postgres image on GHCR</li> </ul>                               | Script tests in CI                            |
| ✓ | <b>M5 WSL distro &amp; ZFS builds</b> | <ul style="list-style-type: none"> <li>✓ Prebuilt distro image</li> <li>✓ Downloaded and verified by the installer (seen on Windows)</li> <li>✓ Preloaded image tagged as the engine runs it</li> </ul> | <ul style="list-style-type: none"> <li>✓ Imported on a fresh Windows install; preloaded images loaded</li> <li>🟡 685 MB download (rootfs fallback covers failures)</li> </ul> | Actions artifacts; Pester; Windows run 16 Sep |
| ✓ | <b>M6 Postgres image</b>              | <ul style="list-style-type: none"> <li>✓ Postgres 16 + wal-g</li> </ul>                                                                                                                                 | <ul style="list-style-type: none"> <li>□ pgvector</li> </ul>                                                                                                                  | Manual                                        |

## N Not started (roadmap)

11 items

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

| ID                         | Item · scope                                                                                                                                         | Related |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <input type="checkbox"/> - | <b>Multi-tenant projects &amp; roles</b><br><input type="checkbox"/> Projects / organisations <input type="checkbox"/> Per-branch permissions (RBAC) | G1, H6  |
| <input type="checkbox"/> - | <b>Branch merge</b><br><input type="checkbox"/> Apply one branch's schema changes to another                                                         | H18     |
| <input type="checkbox"/> - | <b>Scheduled backups &amp; retention</b>                                                                                                             | D2      |
| <input type="checkbox"/> - | <b>Automatic HA failover</b><br><input type="checkbox"/> Health detection, fencing <input type="checkbox"/> Multi-host                               | D4      |
| <input type="checkbox"/> - | <b>Pipelines: scheduling, incremental / CDC extract, DAG view</b>                                                                                    | J1      |
| <input type="checkbox"/> - | <b>More import sources: Cassandra, DynamoDB</b>                                                                                                      | I1–I8   |
| <input type="checkbox"/> - | <b>pgvector in the Postgres image</b>                                                                                                                | M6      |
| <input type="checkbox"/> - | <b>Cloud write-once anchors</b> (S3 Object Lock etc.)                                                                                                | H10     |
| <input type="checkbox"/> - | <b>Release signatures</b>                                                                                                                            | A9, M4  |
| <input type="checkbox"/> - | <b>Hosted cloud tier</b>                                                                                                                             | —       |
| <input type="checkbox"/> - | <b>Networked storage disaggregation</b> (README roadmap; see the storage engine note on page 1)                                                      | C1, D1  |

### Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.